

Samir Sen

AI Robotics & Automation Engineer · Computer Vision · Edge AI

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[LinkedIn](#) | [GitHub](#) | [Portfolio Website](#)

PROFESSIONAL SUMMARY

AI robotics engineer at Baliyo Ventures, building perception and autonomy on NVIDIA Jetson hardware with ROS 2, LiDAR and computer vision. Works across the full stack, from sensor drivers and on-device models to FastAPI and Spring Boot services in Docker. BSc (Hons) in Computing with AI, London Metropolitan University.

TECHNICAL SKILLS

Robotics: ROS 2, SLAM, Nav2, odometry, path planning, LiDAR, RGB-D cameras

AI & Computer Vision: PyTorch, TensorFlow, OpenCV, SSD MobileNet, scikit-learn, Pandas

Edge & Embedded: NVIDIA Jetson (Orin NX, Nano), ESP32, Arduino, serial protocols

Languages: Python, Java, C++, SQL, JavaScript

Backend & Tools: FastAPI, Spring Boot, REST APIs, Docker, Linux, Git, AWS, Azure

EXPERIENCE

R&D AI Robotist & Automation Engineer

Aug 2026 – Present

Baliyo Ventures, Kathmandu

- Building a LiDAR SLAM and autonomous navigation system for a car-like robot, using ROS 2 on an NVIDIA Jetson Orin NX.
- Wrote a custom LiDAR driver in Python with no vendor SDK: 8 Hz scans, ~2,500 points per turn, zero dropped packets.
- Planned a step-by-step autonomy roadmap (odometry, mapping, localization, path planning, exploration), benchmarked against slam_toolbox and Nav2.
- Leading **Bandar Sentinel**, an offline-first edge-AI system that detects monkeys, classifies their behaviour and triggers non-lethal deterrents to protect crops.

PROJECTS

Real-Time Object Detection & Distance Estimation · [GitHub](#) · *SSD MobileNet, OpenCV, FastAPI, Spring Boot, Docker*

- Built a pipeline that detects objects in live video and estimates their distance and direction for robotic decisions.
- Served the model through a FastAPI inference service behind a Spring Boot web app with user login, deployed with Docker.

Spatial Edge Intelligence · *Jetson Nano, LiDAR, PyTorch*

- Combined LiDAR and depth sensing with on-device PyTorch inference, so the robot reasons about its surroundings without remote computing.

IoT Environmental Monitor · [GitHub](#) · *ESP32, DHT11, GSM, C++*

- Sent live temperature and humidity readings to the cloud and by SMS for sites without reliable internet.

Weather Prediction · [GitHub](#) · *Python, scikit-learn, Random Forest*

- Built forecasting models with time-series feature engineering, comparing Random Forest and linear regression by RMSE and MAE.

Inventory Management API · [GitHub](#) · *Java, Spring Boot, JWT, MySQL*

- Built a REST backend with JWT authentication, role-based access and CSV import/export.

EDUCATION

BSc (Hons) Computing with Artificial Intelligence

2022 – 2025

Islington College (London Metropolitan University), Kathmandu

- Coursework: machine learning, neural networks, NLP, big data, algorithms, cloud computing.
- Final-year project: real-time object detection and distance estimation (see Projects).

CERTIFICATIONS & ACTIVITIES

- **AWS Cloud Practitioner** (Cloud Quest) 2023
- **50+ public repositories** on GitHub, including open robotics research logs
- **Club Tamer**, Leo Club of Temple Peace Kavre 2021 – 2022